SOLUTION
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (A)

2. Ans. (B)

Sol. In rod, $\frac{\lambda}{4} = \ell \Rightarrow \lambda = 4\ell \Rightarrow v_r = \lambda\nu = 4\ell\nu$

In air column $v_a = 2\Delta\ell\nu$; $\frac{v_r}{v_a} = \frac{2\ell}{\Delta\ell} \Rightarrow v_r = \frac{2\times1\times330}{0.6} = 1100 m/s$

3. Ans. (B)

Sol. Let n_1 be the frequency of the wire having tension T + ΔT and n_2 be the frequency of the wire having tension T, then $\frac{n_1}{n_2} = \sqrt{\frac{T+\Delta T}{T}} = \frac{500+5}{500} = \frac{505}{500}$

$$\Rightarrow \frac{T+\Delta T}{T} = \left(1+\frac{5}{500}\right)^2 \approx 1 + \frac{2}{100} \approx 1.02 \Rightarrow \frac{\Delta T}{T} = 0.02$$

OR

For stretched wire frequency $n \propto \sqrt{T} \Rightarrow \frac{\Delta n}{n} = \frac{1}{2}\frac{\Delta T}{T}$

$$\Rightarrow \frac{\Delta T}{T} = 2\left(\frac{\Delta n}{n}\right) = 2\left(\frac{5}{500}\right) = 0.02$$

4. Ans. (C)

Sol. The second wire resonates best with the first wire when the two wires have **exactly** the same frequency.

$$f = \frac{n}{2\ell}\sqrt{\frac{T}{\rho A}} = \frac{n}{2\ell}\sqrt{\frac{T}{\rho\pi r^2}} \Rightarrow n \propto r$$

5. Ans. (D)

Sol. At closed end pressure antinode and displacement nodes are formed so at closed end variation in pressure is maximum.

6. Ans. (A)

Sol. Loudness $\beta = 10\log_{10}\frac{I}{I_0}$ $\therefore \beta_2 - \beta_1 = 10\log_{10}\frac{I_2}{I_1}$ $\because I = \frac{P}{4\pi r^2}$ $\therefore \frac{I_2}{I_1} = \frac{r_1^2}{r_2^2}$

$$\therefore (\beta + 20) - \beta = 10\log_{10}\frac{r^2}{r_2^2} = 20\log_{10}\frac{r}{r_2} \Rightarrow \frac{r}{r_2} = 10 \Rightarrow r_2 = 0.1r \quad \therefore \text{shift} = r - 0.1\,r = 0.9\,r.$$

7. Ans. (C)

Sol. Appearent frequency $f = \left(\frac{v}{v - u\cos\theta}\right)f_0$

Angle of approaches θ,

$$\theta_C > \theta_B > \theta_A$$

$\therefore$ $$f_1 > f_2 > f_3$$

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8. Ans. (C)

Sol. Wave originating at t =0 from S_1 reaches point A at t = 1.

Wave originating at $t = \frac{1}{2}$ from S_2 reaches point A at t = 1.

So phase difference in these waves = $\frac{\pi}{2}+\frac{\pi}{6}$

$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi} = 5$

9. Ans. (C)

Sol. $\Delta\phi = 2n\pi \Rightarrow \frac{\pi}{2} + \frac{2\pi}{\lambda} d\sin\theta = 2n\pi;\ \frac{2\pi}{\lambda} d\sin\theta = \left(2n-\frac{1}{2}\right)\pi$

$\sin\theta = \left(2n-\frac{1}{2}\right)\ \frac{\lambda}{2d} = \frac{1}{2}\times\frac{\lambda}{2\times3\lambda} = \frac{1}{12} \Rightarrow \frac{y}{\sqrt{(100\lambda)^2}} = \frac{1}{12}$

$144y^2 = (100\lambda)^2\ ;\ y \approx \frac{100\lambda}{12} = \frac{25\lambda}{3}$

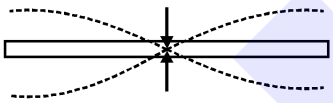
Multiple Correct Answer Type — 8 Q. [4 M (–1)]

10. Ans. (A, C)

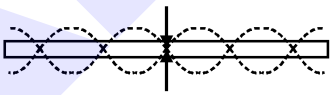
11. Ans. (A, B, C, D)

12. Ans. (A,C)

Sol. $L = \frac{\lambda}{2} \Rightarrow \lambda = 2L$



$\Rightarrow\ f = \frac{C}{2L}$

2nd OT $\Rightarrow L = \frac{5\lambda}{2}$ 

$\Rightarrow\ \lambda = \frac{2L}{5}\ ;\ f = \frac{5C}{2L}$

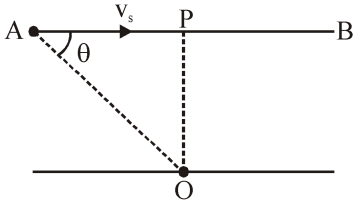
f is $\frac{C}{2L}, \frac{3C}{2L}, \frac{5C}{2L}$ etc.

13. Ans. (C,D)

Sol. The graph shows the situation shown in figure below. The observed frequency will initially be more than the natural frequency. When the source is at P, observed frequency is equal to its natural frequency i.e., 2000 Hz.

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For region AP : $f = f_0\left(\frac{v}{v - v_s\cos\theta}\right)$



For PB : $f = f_0\left(\frac{v}{v + v_s\cos\theta}\right)$

Minimum value of f will be:

$$f_{min} = f_0\left(\frac{v}{v + v_s}\right) \text{ when } \cos\theta = 1$$

or $1800 = 2000\left(\frac{300}{300 + v_s}\right)$

Solving this we get, $v_s = 33.33$ m/s
and maximum value of f can be

$$f_{max} = f_0\left(\frac{v}{v - v_s}\right) \text{ when } \cos\theta = 1$$

or $f_{max} = 2000\left(\frac{300}{300 - 33.33}\right) = 2250$ Hz

14. Ans. (A,C)

Sol. For sound, air is denser medium than water.

15. Ans. (B,D)

Sol. From open end compression will be reflected as refraction and from closed end these is no phase change so compression will be reflected back as compression.

16. Ans. (A, B, C)

Sol. For open organ pipe $f = \frac{nv}{2L}$ & for closed $f = \frac{(2n+1)}{4L}$

$$\frac{v}{2(24+22)} = \frac{3v}{4(38+e)} \Rightarrow e = 1 \text{ cm}$$

Fundamental of open pipe = $\frac{340}{2(26\text{cm})} = 654$ Hz

17. Ans. (A,B, D)

Sol. (A) $y_1 = A\sin 2\pi ft$ $\quad y_2 = y_1 + y_2$

$y_2 = A\sin 2\pi f_2 t$

$f = \frac{f_1 - f_2}{2}$

for pressure wave it behave like open end.

Linked Comprehension Type (1 Para × 4Q.) (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

18. Ans. (A)

Sol. In first case, no relative motion between source and observer so no doppler effect. Only due to distance, last soldiers heard sound after some time.

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19. Ans. (B)

Sol. time interval between strike left foot and next strike right foot = $\frac{60\,\text{sec}}{120} = \frac{1}{2}\text{sec}.$

time interval between first and last soldier is = $n\left(\frac{1}{2}\text{sec}.\right)$

n is integer
distance between first and last soldier S = vt

$$S = (320)\,(n \times \frac{1}{2}) = 160\text{m}$$

$$S = 160, 320, 480,.....$$

20. Ans. (D)

Sol. In part b, soldiers recieve sound of frequency $\frac{121}{60}$Hz

$$\Rightarrow f' = \left(\frac{v + v_0}{v}\right) f$$

$$\Rightarrow \frac{121}{60} = \left(\frac{320 + v_0}{320}\right)\frac{120}{60} \Rightarrow 320 \times 121 = 320 \times 120 + 120\, v$$

$$\Rightarrow v_0 = \frac{320}{120} = 2.66 \text{ m/s}$$

21. Ans. (B)

Sol. $f' = \left(\frac{v - v_0}{v}\right) f = \left(\frac{320 - \frac{320}{120}}{320}\right) \cdot 120 = 119$ steps/min.

22. Ans. (A)

Sol. $f = \frac{C - V_R}{C + V_R} f_0 ; \frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R} \approx \frac{2V_R}{C} \Rightarrow C = \frac{2 \times 0.1}{600} \times 5 \times 10^6 \approx 1700$ m/s

23. Ans. (D)

Sol. $f' = \frac{C - V_R}{C + V_R} f_0 ; \frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R}$

velocity is 4 times ($A_1 v_1 = A_2 v_2$) $\Rightarrow \Delta f = 4\Delta f_0 = 2400$ Hz

24. Ans. (A)

Sol. $\frac{1}{2}\frac{dm}{dt}(v_2^2 - v_1^2) = \frac{1}{2} \times 1.5 \times 10^3 \times 0.1 \times 10^{-4} \times 0.1\, [(4 \times 0.1)^2 - (0.1)^2]$

$= 0.75 \times 10^{-5}\, [15] = 11.25 \times 10^{-5} = 1.125 \times 10^{-4}$ W

SECTION-II

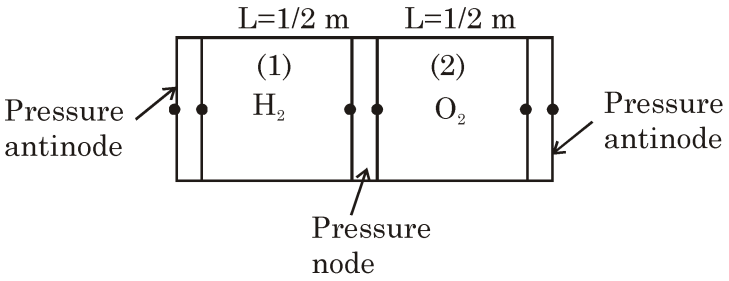
Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

25. Ans. 1650 Hz

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Sol.



$$n_1 \times \frac{v_1}{4L} = n_2 \times \frac{v_2}{4L}$$

$$\frac{n_1}{n_2} = \frac{300}{1100} = \frac{3}{11}$$

$$f_1 = \frac{3 \times 1200}{4 \times \frac{1}{2}} = 1650\text{Hz}$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

26. Ans. 2

Sol. $f_{reflected} = \frac{(300-150)}{150} \times 75\text{Hz}$

$\therefore \lambda_{reflected} = 2$ m

27. Ans. 4

Sol. $f = \frac{v}{2(\ell + 2e)}$ where e = end correction = 0.6 r

$$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)} \therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r} = \frac{\Delta v}{v} - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r}$$

here $\frac{\Delta v}{v} = 0$ (given) $\frac{\Delta f}{f} \times 100 = -\frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r} \times 100$

for maximum % error : $\Delta\ell = 0.1$, $\Delta r = 0.05$

$$\left(\frac{\Delta f}{f} \times 100\right)_{max} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100 = \mathbf{0.16\%}$$

28. Ans. 2

Sol. $I = \frac{P^2}{2\rho V}$, $v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{4 \times 10^4}{1}} = 200$ m/s Also $20 = 10 \log \frac{I}{I_0} \Rightarrow I = 10^{-19}$ W/m^2

$$10^{-10} = \frac{P^2}{2 \times \rho \times v} \Rightarrow P = 2 \times 10^{-4} \text{ N/m}^2$$

29. Ans. 4

Sol. Energy flow rate reaching surface of the ball = $\pi R^2 \frac{(\Delta p_m)^2}{2\rho v \lambda}$

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30. Ans. 2

Sol. Let be ℓ is wavelength of sound in air

$$\ell_1 = \frac{\lambda}{4} \text{ and } \ell_2 = \frac{3\lambda}{4}$$

$$\lambda = 2(\ell_2 - \ell_1)$$

$$v = f\lambda = 165 \times 2 (150 - 50) \times 10^{-2} = 330 \text{ m/s}$$

$$\frac{dv}{v} = \frac{|d\ell_2| + |d\ell_1|}{\ell_2 - \ell_1}$$

$$dv = \frac{0.5 + 0.1}{150 - 50} \times 330 = 1.98 \text{m/s}$$

Subjective Type **1 Q. [4 M (0)]**

31. Ans. (a) $r = \sqrt{\frac{250}{\pi}}$ m, (b) $-\frac{4A}{55}$

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